MA1522 Linear Algebra for Computing

Lesson 9: Subspaces and Linear Independence

Announcement: Online Quiz 2

- **Quiz 2 covers Chapter 2**, Sections 2.1–2.10: matrix algebra, matrix equations, inverses and elementary matrices, LU factorization, and determinants.

- It opens **Saturday 12 September, 0:00** and is due **Sunday 13 September, 23:59**.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson9` (e.g. `A1234567Z_lesson9.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 9**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

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Example

Let V and W be subspaces of $\mathbb{R}^n$. True or false?

1. $V \cup W$ is a subspace of $\mathbb{R}^n$.
2. $V \cap W$ is a subspace of $\mathbb{R}^n$.

(1) **False.** In $\mathbb{R}^3$ take $V = \{(x, y, z)^T \mid x = 0\}$ and $W = \{(x, y, z)^T \mid y = 0\}$, both subspaces. Then

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \in V, \qquad \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \in W, \qquad \text{but} \qquad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \notin V \cup W.$$

Here $V \cup W = \{(x, y, z)^T \mid xy = 0\}$, which is last lesson's V_3: it failed only closure under addition, and this is the reason.

(2) **True.** $\mathbf{0} \in V$ and $\mathbf{0} \in W$, so $\mathbf{0} \in V \cap W$. Let $\mathbf{v}_1, \mathbf{v}_2 \in V \cap W$ and $\alpha \in \mathbb{R}$. Then $\mathbf{v}_1, \mathbf{v}_2$ lie in V, so $\mathbf{v}_1 + \mathbf{v}_2$ and $\alpha\mathbf{v}_1$ lie in V; the same argument in W puts them in W. Hence both lie in $V \cap W$.

Section 3.4 Computing $V \cap W$

Let $V = \text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ and $W = \text{span}\{\mathbf{w}_1, \ldots, \mathbf{w}_l\}$ be subspaces of $\mathbb{R}^n$. Then

$$\begin{aligned}
\mathbf{v} \in V \cap W &\iff \mathbf{v} \in V \text{ and } \mathbf{v} \in W \\
&\iff \mathbf{v} = c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k = d_1\mathbf{w}_1 + \cdots + d_l\mathbf{w}_l \quad \text{for some } c_i, d_j \in \mathbb{R} \\
&\iff \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_k & \mathbf{w}_1 & \cdots & \mathbf{w}_l \end{pmatrix} \begin{pmatrix} c_1, & \ldots, & c_k, & -d_1, & \ldots, & -d_l \end{pmatrix}^T = \mathbf{0}.
\end{aligned}$$

So: solve that homogeneous system, then read the first k entries of each solution and substitute back,

$$\mathbf{v} = c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k.$$

Question 1

Let $V = \text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$ and $W = \text{span}\{\mathbf{w}_1, \mathbf{w}_2\}$ be subspaces of $\mathbb{R}^3$, where

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}, \quad \mathbf{w}_1 = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

Compute $V \cap W$.

Question 1 — solution

Stand the four vectors up as columns, in the order $\mathbf{u}_1, \mathbf{u}_2, \mathbf{w}_1, \mathbf{w}_2$, and reduce.

$$\begin{pmatrix} 1 & 3 & 2 & 2 \\ 1 & 2 & 2 & 1 \\ 2 & 2 & 3 & 1 \end{pmatrix} \xrightarrow{R_2 - R_1,\ R_3 - 2R_1} \begin{pmatrix} 1 & 3 & 2 & 2 \\ 0 & -1 & 0 & -1 \\ 0 & -4 & -1 & -3 \end{pmatrix} \xrightarrow{R_3 - 4R_2} \begin{pmatrix} 1 & 3 & 2 & 2 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}.$$

Column 4 is the only non-pivot column. Setting $-d_2 = t$ and working back up, $-d_1 = t$, $c_2 = -t$ and $c_1 = -t$, so

$$(c_1,\ c_2,\ -d_1,\ -d_2)^T = t\,(-1, -1, 1, 1)^T, \qquad t \in \mathbb{R}.$$

We conclude that $\mathbf{v}$ is in the intersection if and only if

$$\mathbf{v} = -t\,\mathbf{u}_1 - t\,\mathbf{u}_2 = -t\,\mathbf{w}_1 - t\,\mathbf{w}_2 \quad \checkmark$$

for some $t \in \mathbb{R}$. Therefore $V \cap W = \operatorname{span}\{\mathbf{u}_1 + \mathbf{u}_2\} = \operatorname{span}\{\mathbf{w}_1 + \mathbf{w}_2\} = \operatorname{span}\{(4, 3, 4)^T\}$. $\square$

Challenge

Let V and W be subspaces of $\mathbb{R}^n$. Show that

$$V \cup W \text{ is a subspace of } \mathbb{R}^n \iff V \subseteq W \text{ or } W \subseteq V.$$

($\Leftarrow$) If $W \subseteq V$ then $V \cup W = V$, which is a subspace; and if $V \subseteq W$ then $V \cup W = W$.

($\Rightarrow$) Suppose $V \cup W$ is a subspace and that neither contains the other. Then there are vectors $\mathbf{v} \in V$ with $\mathbf{v} \notin W$, and $\mathbf{w} \in W$ with $\mathbf{w} \notin V$. Both lie in $V \cup W$, so $\mathbf{v} + \mathbf{w} \in V \cup W$: that is, $\mathbf{v} + \mathbf{w} \in V$ or $\mathbf{v} + \mathbf{w} \in W$.

If $\mathbf{v} + \mathbf{w} \in V$, then $\mathbf{w} = (\mathbf{v} + \mathbf{w}) + (-1)\mathbf{v} \in V$, since V is closed under scalar multiplication and addition — contradicting $\mathbf{w} \notin V$. If $\mathbf{v} + \mathbf{w} \in W$, the same argument gives $\mathbf{v} = (\mathbf{v} + \mathbf{w}) + (-1)\mathbf{w} \in W$, contradicting $\mathbf{v} \notin W$. So one of V, W contains the other. $\square$

Section 3.5 Linear Independence

Definition

A set $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ is **linearly independent** if the only coefficients $c_1, c_2, \ldots, c_k$ satisfying

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_k\mathbf{u}_k = \mathbf{0}$$

are $c_1 = c_2 = \cdots = c_k = 0$. Otherwise the set is **linearly dependent**.

A set is linearly dependent exactly when one of its vectors is a linear combination of the others: if $c_i \neq 0$ in such a relation, then

$$\mathbf{u}_i = \frac{c_1}{-c_i}\mathbf{u}_1 + \cdots + \frac{c_{i-1}}{-c_i}\mathbf{u}_{i-1} + \frac{c_{i+1}}{-c_i}\mathbf{u}_{i+1} + \cdots + \frac{c_k}{-c_i}\mathbf{u}_k.$$

Section 3.5 Checking Linear Independence

Theorem

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ be a set of vectors in $\mathbb{R}^n$ and let $A = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k \end{pmatrix}$. The following are equivalent:

(i) S is linearly independent;

(ii) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution;

(iii) the reduced row-echelon form of A has no non-pivot column.

Three special cases:

1. $\{\mathbf{0}\}$ is linearly dependent.
2. $\{\mathbf{v}\}$ is linearly independent if and only if $\mathbf{v} \neq \mathbf{0}$.
3. $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly dependent if and only if one is a scalar multiple of the other.

Question 2

For each of the following sets, determine whether it is linearly independent. If it is not, exhibit a linearly independent subset with the same linear span.

1. $S_1 = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} \right\}$

2. $S_2 = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix}, \quad \mathbf{v}_4 = \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}.$$

Question 2 — solution: S_1

Stand the three vectors up as columns and reduce.

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{pmatrix} \xrightarrow{R_2 - 2R_1,\ R_3 - 3R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -5 \\ 0 & -5 & -7 \end{pmatrix} \xrightarrow{R_3 - 5R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -5 \\ 0 & 0 & 18 \end{pmatrix}.$$

This is already in row-echelon form and all three columns are pivot columns, so the homogeneous system has only the trivial solution and S_1 is **linearly independent.** There is no relation to write down and no smaller subset to find.

Question 2 — solution: S_2

Stand the four vectors up as columns and reduce.

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 4 \\ -1 & 1 & -3 & 3 \end{pmatrix} \xrightarrow{R_2-2R_1,\ R_3+R_1} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 2 & -2 & 4 \end{pmatrix} \xrightarrow[R_1-R_2]{R_3-2R_2} \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Columns 3 and 4 are non-pivot, so S_2 is **linearly dependent**. Each non-pivot column lists the coefficients of that vector in terms of the pivot columns:

$$\mathbf{v}_3 = 2\mathbf{v}_1 - \mathbf{v}_2, \quad \checkmark \qquad \mathbf{v}_4 = -\mathbf{v}_1 + 2\mathbf{v}_2.$$

So $\mathbf{v}_3$ and $\mathbf{v}_4$ lie in $\operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ and $\operatorname{span}(S_2) = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$, where $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent because neither vector is a multiple of the other. □

Question 3

Let $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ be a linearly independent set of vectors in $\mathbb{R}^n$. Determine whether each of the following sets is linearly independent.

1. $S = \{\ \mathbf{u} - \mathbf{v},\ \ \mathbf{v} - \mathbf{w},\ \ \mathbf{w} - \mathbf{u}\ \}$

2. $T = \{\ \mathbf{u} + \mathbf{v},\ \ \mathbf{v} + \mathbf{w},\ \ \mathbf{w} + \mathbf{u}\ \}$

Question 3 — solution: S

Take a general relation and collect the coefficients of $\mathbf{u}, \mathbf{v}, \mathbf{w}$:

$$c_1(\mathbf{u}-\mathbf{v}) + c_2(\mathbf{v}-\mathbf{w}) + c_3(\mathbf{w}-\mathbf{u}) = (c_1 - c_3)\mathbf{u} + (c_2 - c_1)\mathbf{v} + (c_3 - c_2)\mathbf{w} = \mathbf{0}.$$

Since $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly independent, each collected coefficient is zero:

$$c_1 - c_3 = 0, \qquad c_2 - c_1 = 0, \qquad c_3 - c_2 = 0 \quad \Longrightarrow \quad c_1 = c_2 = c_3.$$

So $c_1 = c_2 = c_3 = 1$ is a relation, and S is **linearly dependent**:

$$(\mathbf{u}-\mathbf{v}) + (\mathbf{v}-\mathbf{w}) + (\mathbf{w}-\mathbf{u}) = \mathbf{0}. \quad \checkmark$$

Question 3 — solution: T

The same first two lines, with the minus signs turned into plus signs:

$$c_1(\mathbf{u}+\mathbf{v})+c_2(\mathbf{v}+\mathbf{w})+c_3(\mathbf{w}+\mathbf{u})=(c_1+c_3)\mathbf{u}+(c_1+c_2)\mathbf{v}+(c_2+c_3)\mathbf{w}=\mathbf{0},$$

so

$$c_1+c_3=0, \qquad c_1+c_2=0, \qquad c_2+c_3=0.$$

The first two give $c_3=-c_1$ and $c_2=-c_1$, and substituting into the third gives $-2c_1=0$. Hence $c_1=0$ and then $c_2=c_3=0$: the only relation is the trivial one, and T is **linearly independent**.

The two sets differ only in three signs, and the answers differ. □

Next lesson

- **Sections 3.6 & 3.7 Basis and Coordinates, Dimensions**

Please watch the video clips for these sections beforehand.